

Welcome to Course 3: Unsupervised Learning, Recommenders, and Reinforcement Learning

This is the third and final course of the Machine Learning specialization. While the previous two courses focused heavily on **Supervised Learning** (where the data has labels), this course introduces a new set of powerful tools to expand your capabilities beyond labeled data.

Course Overview

Week 1: Unsupervised Learning

This week focuses on finding patterns in data that does not have specific output labels.

- **Clustering:** Algorithms that automatically group data into coherent clusters.
- **Anomaly Detection:** Techniques used to identify unusual data points (crucial for fraud detection and system monitoring).

Week 2: Recommender Systems

You will learn the logic behind the engines that drive modern online commerce and media.

- **Applications:** How websites like Amazon or Netflix decide which products or movies to suggest to you.
- **Importance:** This is one of the most commercially valuable technologies in ML today, responsible for moving billions of dollars in value, despite receiving less attention in academia than other topics.
- **Ad Tech:** These concepts also apply to how online advertising systems decide which ads to show a user.

Week 3: Reinforcement Learning (RL)

The course concludes with an exciting, emerging frontier of AI.

- **Concept:** Teaching an agent to behave in an environment to maximize a reward.
- **Applications:** Used to train computers to play video games (often outperforming humans) and control robots.
- **Project:** You will implement an RL algorithm to land a simulated **moon lander**.
- **Status:** While currently having fewer commercial applications than Supervised Learning, RL is opening new possibilities for what algorithms can achieve.